## Future Directions I

1. **Supply-chain provenance**: Integration with software supply chain frameworks such as in-toto (Torres-Arias et al. 2019) and SLSA (Supply-chain Levels for Software Artifacts) (Open Source Security Foundation 2023) to provide end-to-end attestation from source commit through build pipeline to published artifact. SLSA's four graduated levels of build integrity (from basic provenance to hermetic, reproducible builds) provide a natural extension ladder for the template's currently document-centric provenance model. The template's existing steganographic layer embeds document-level provenance; supply-chain frameworks would add build-level provenance, closing the gap between "this PDF was produced by this pipeline" and "this pipeline was executed with this verified source code."
2. **Decentralized provenance**: Integration with IPFS or Arweave for immutable publication records, extending the SHA-256-based tamper detection to content-addressed storage networks.
3. **Digital signatures**: GPG or X.509 signing integrated into the steganographic layer, providing cryptographic non-repudiation in addition to tamper detection.
4. **Continuous integration**: GitHub Actions workflows that execute the pipeline on every push, with PDF artifacts as release assets and automated DOI registration via Zenodo.

# Future Directions II

5. **Multi-language support**: Extension of the Thin Orchestrator pattern to R, Julia, and Rust projects, enabling polyglot research workflows within the Two-Layer Architecture.
6. **Automated FAIR4RS assessment**: Periodic self-scoring against FAIRsoft metrics (Garijo et al. 2024), with quality indicators (executability, documentation completeness, metadata richness) tracked as pipeline artifacts alongside test coverage and rendering status.
7. **Knowledge graph integration**: Connecting pipeline outputs to Active Inference Knowledge Base entries for automated meta-analysis and cross-project citation tracking.
8. **Formal verification**: Static analysis tooling to enforce the Thin Orchestrator pattern—verifying that scripts contain no algorithmic logic and that `src/` modules do not import from `scripts/`.

## Future Directions III

9. **Agentic research pipelines via MCP**: The `SKILL.md` descriptors already define the interface contracts for each infrastructure module; the natural next step is to expose them as MCP server endpoints (Anthropic 2024). An MCP server wrapping `infrastructure.llm` would expose `query`, `review_manuscript`, and `translate_abstract` as protocol-native Tools; an MCP server wrapping `infrastructure.publishing` would expose `publish_to_zenodo` and `generate_citation_bibtex`. A research agent could then compose these Tools to execute the full pipeline—environment setup → test execution → analysis → rendering → validation → LLM review → DOI registration—without any human in the loop. This closes the loop opened by Lu et al.'s AI Scientist (Lu et al. 2024), which demonstrated automated hypothesis generation and experimental iteration but relied on ad hoc laboratory scaffolding. `template/`'s pipeline, fully exposed as MCP tools, provides that scaffolding in a reproducible, versioned, and certified form. Longer-term, the Agent2Agent (A2A) protocol (Google 2025) enables heterogeneous specialist agents—a statistical analyst, a figure designer, a peer-review simulator—to coordinate via a shared, protocol-mediated workspace built on precisely the kind of modular, well-documented infrastructure that `template/` provides.

10. **Research Infrastructure as Code and software citation**: The DevOps IaC paradigm, applied to research, means the entire manuscript pipeline is a version-controlled, reproducible artifact in its own right. Software Heritage (Di Cosmo et al. 2020) provides persistent SWHIDs (Software Hash Identifiers) for source-code snapshots, enabling `template/` itself to be cited as a software artifact with a DOI-equivalent stable identifier. Combining this with Zenodo DOI registration (already supported by `infrastructure.publishing`) creates a full citation chain: the paper cites the data (DOI), the data provenance cites the pipeline (SWHID), and the pipeline cites the framework release (Zenodo DOI). This three-link citation chain operationalizes the Katz et al. (Katz et al. 2021) software citation principles at the infrastructure level.

## Conclusion I

`template/` demonstrates that high-integrity, reproducible research need not be onerous. By embedding provenance, testing, and documentation into the architecture itself—rather than layering them atop a fragmented workflow—the template transforms "best practices" from aspirational guidelines into enforced invariants (Wilson et al. 2017; Sandve et al. 2013; Lamprecht et al. 2020). The Two-Layer Architecture ensures that infrastructure improvements propagate to all projects without coupling. The Zero-Mock policy ensures that tests reflect reality. The steganographic provenance layer ensures that published artifacts carry their own authentication. The comparative analysis distinguishes native, partial, and absent support across its versioned

capability inventory. The total comparison inventory includes capabilities that `template/` does not fully support, such as distributed execution and multi-language research. It is not the count of fully native `template/` capabilities, and the snapshot does not establish an exhaustive claim about all existing tools.

The template is not merely a build tool; it is an epistemological commitment. It asserts that a research paper is not a static document but a build artifact—reproducible, verifiable, and traceable to the code that generated it. As Knuth observed, programs should be written for humans to read and only incidentally for machines to execute (Knuth 1984). We extend this dictum: research manuscripts should be *built* for verification and only incidentally for reading. In an era of generative AI, AI-native research workspaces, and synthetic media—where the boundary between human-authored and machine-generated

text grows increasingly indeterminate (Gruenpeter et al. 2021)—the provenance chain from source code to published PDF is not an administrative convenience. It is the epistemic ground on which scientific trust must be rebuilt. That this manuscript was itself built, tested, and watermarked by the pipeline it describes—its metrics computed from the repository it inhabits, its figures rendered by the code it documents—demonstrates one reproducible artifact path; broader reliability claims require the declared checks and candidate-bound execution evidence.